Anirudh Verma

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Education

Indian Institute of Science, Bengaluru (IISc) 2020 – 2025
Physics Major, Bachelor and Master of Science (Research). CGPA: 8.5/10 (BS), 8.4/10 (BS - MS)

Holy Wisdom School, Champawat, Uttarakhand 2020
High School (AISSCE): 97.6%

Udayan International School, Champawat, Uttarakhand 2018
Middle School (AISSE): 94.2%

Publications

A Quantum-Walk Representation of Color-Ordered MHV Scattering Amplitudes 2026
Anirudh Verma, C. M. Chandrashekar
(*Under review at Physical Review D*):arXiv:2607.02456

Photonic Simulation of Beyond-Quantum Nonlocal Correlations (e.g. Popescu-Rohrlich Box) with Non-Signaling Quantum Resources 2025
Kunal Shukla, Anirudh Verma, Kanad Sengupta, Sanchari Chakraborti, Manik Banik, C. M. Chandrashekar
(*Under review at NPJ Quantum Information*):arXiv:2509.26271

Four-Qubit Photonic System for Publicly Verifiable Quantum Random Numbers and Generation of Public and Private Key 2024
Mayalakshmi Kolangatt, Anirudh Verma, Sujai Matta, Kanad Sengupta, C. M. Chandrashekar
Physical Review A 110, 032615

Bachelor and Master Thesis

Entanglement Certified Quantum Random Number Generation using Polarisation Entangled Photons 2023–24
Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore (Bachelor Thesis Advisor)

Response and Correlations of Weakly Measured Free Fermions 2024–25
Supervisor: Prof. Sumilan Banerjee, Associate Professor, IISc Bangalore (Master Thesis Advisor)

Research Projects

Quantum Walk Dynamics and PR Correlations (Project Assistant) June 2025 – Present
Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore

- Currently working on quantum tokenized signature scheme.
- Simulated **post-quantum nonlocal correlations**, such as Popescu–Rohrlich boxes, using intrinsically non-signaling resources implemented on a four-qubit photonic platform.
- Simulated **quantum walks in spin Hamiltonians** (e.g., XXZ, TFIM) to understand the dynamics in a lattice configuration.
- Investigated scattering amplitudes using **quantum walks**.

Quantum Simulation using Variational and Sample-Based Methods Sep 2025 – Dec 2025
Supervisor: Prof. Viki Kumar Prasad, University of Calgary

- Built a quantum chemistry simulation pipeline using **PySCF** and **Qiskit Nature** with ACP-corrected Hamiltonians for noncovalent systems.
- Performed **active space reduction** and integral projection to enable efficient quantum circuit simulation.
- Applied **VQE** with UCCSD ansatz and benchmarked against SCF and exact diagonalization.
- Implemented **Sample-Based Quantum Diagonalization (SQD)** using subspace projection from parameterized quantum states.

Linear Response Theory across Measurement-Induced Entanglement Transition Sep 2024 – Apr 2025
Supervisor: Prof. Sumilan Banerjee, Associate Professor, Department of Physics, IISc Bangalore

- Worked in the field of **open quantum many-body systems**, studying **measurement-induced entanglement transitions** in **free fermionic chains** under continuous weak monitoring.
- Used **stochastic Schrödinger equations** to simulate quantum trajectories and explore the interplay of **unitary dynamics** with **local number measurements**.

- Benchmarked a crossover from **logarithmic** to **area-law** entanglement scaling and analyzed **density-density** and **current-current** correlation decay.
- **Term Paper:** *Simulation of Quantum Trajectories and Measurement-Induced Entanglement Transition*

Quantum Random Number Generation using Polarisation Entangled Photons Sep 2023 – Apr 2024
Supervisor: Prof. C. M. Chandrashekar, Associate Professor, IISc Bangalore

- Developed a **polarisation-entangled photon source** via **Type-II SPDC** using a PPKTP crystal, achieving high-fidelity entanglement verified through **Bell-CHSH inequality** and **quantum state tomography**.
- Demonstrated protocols for **publicly verifiable QRNG**, **entanglement verification**, and simultaneous generation of **public/private cryptographic keys**.
- Generated **secure random bits at 185 kbps**, with outputs passing all **15 NIST randomness tests** without post-processing.

Quantum Optics: Mondal Lab Aug 2023 – Oct 2023
Supervisor: Prof. Partha Pratim Mondal, Professor, IISc Bangalore

- Studied super-resolution optical microscopy, specifically **Single Molecule Localisation Microscopy**, and methods for improving spatial resolution beyond diffraction-limited techniques.

Diagnostics, Control and Data Processing for the STOR-M Tokamak Experiments May 2023 – July 2023
Supervisor: Prof. Chijin Xiao, Professor, University of Saskatchewan, Canada

- Developed an interactive data-analysis code for STOR-M Tokamak plasma discharges across different oscilloscopes; modified legacy MDSPlus archiving programs (MATLAB) and extended a Python GUI for fluctuation-signal analysis.

Machine Learning Projects

Self-directed projects building applied experience across supervised and unsupervised learning, deep learning, and LLM applications. Code: github.com/anirudhverma33/ml-projects

Dialogue Summarisation with a Fine-Tuned Transformer 2026
Personal project • GitHub

- Fine-tuned **T5-small** on the **SAMSum** dialogue dataset with the **Hugging Face Trainer** (PyTorch, CUDA); validation loss decreased from 0.381 to 0.350 over 6 epochs.
- Deployed the model as a **FastAPI** web app with a REST endpoint and beam-search decoding.

Retrieval-Augmented Generation (RAG) Pipeline 2026
Personal project • GitHub

- Built a document question-answering pipeline with **LangChain**: PDF ingestion, recursive chunking, **sentence-transformer** embeddings and a persistent **ChromaDB** vector store.
- Implemented top-k semantic retrieval and context-grounded answer generation with an LLM served through the **Groq** API.

Loan Approval Prediction: End-to-End Supervised ML Pipeline 2026
Personal project

- Built a complete binary classification pipeline to predict loan approval, covering exploratory data analysis, imputation, encoding, scaling, feature engineering, model training, and evaluation.
- Compared **K-Nearest Neighbors**, **Logistic Regression**, and **Naive Bayes**; after feature engineering, logistic regression reached accuracy 0.88 and F1 0.81 on the held-out set.

Deep Learning and Classical ML 2026
Personal projects

- **PyTorch** neural networks: ANN classifier (95.6% test accuracy) and regressor ($R^2 = 0.935$), a **CNN** on CIFAR-10 (72.5%), and an **RNN** for IMDB sentiment analysis (85.2%).
- Ensemble models for health-risk classification (**Random Forest**, 93.8% accuracy), pruned decision trees, SVMs, clustering (K-means, DBSCAN), PCA and anomaly detection.
- Implemented KNN, linear regression and logistic regression **from scratch in NumPy**.

Relevant Courses

- Classical and Quantum Optics
- Condensed Matter Physics I
- Non-Linear and Ultrafast Optics
- Discrete Photonics and Quantum Analogies
- Algorithms and Programming
- Introduction to Fourier Optics

- Introduction to Quantum Measurement
- Introduction to Quantum Computation
- Quantum Mechanics II
- Analysis and Linear Algebra II
- Statistical Mechanics
- Electromagnetic Theory
- Non-Equilibrium Quantum Many-Body Dynamics

Technical Skills

- **Programming:** Python, SQL, MATLAB, Mathematica, HTML, CSS
- **Scientific Libraries:** NumPy, SciPy, SymPy, Pandas, Matplotlib, Seaborn, Qiskit, QuTiP, QuSpin, TeNPy
- **Machine Learning:** PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, LangChain, Sentence-Transformers, ChromaDB
- **Software and Tools:** LaTeX, Git/GitHub, FastAPI, Flask, Jupyter, SciDAVis, WordPress
- **Operating Systems:** Linux, Windows
- **Languages:** English, Hindi

Awards and Honors

- Mitacs* Globalink Research Intern, 2023
- Honoured by the District Magistrate for securing 2nd rank in AISSCE 2019–20 at the district level
- Qualified JEE Mains & JEE Advanced in the top 1 percentile, 2020
- IISc Promotional Scheme Scholar, Nov 2020 – Apr 2025
- IBM Quantum Challenge Badge, 2024
- IBM Quantum Excellence Badge, 2025

*Mitacs Globalink Research Internship is a competitive initiative for international undergraduates to participate in a 12-week research internship under the supervision of Canadian university faculty members across science, engineering, mathematics, and the humanities.

Workshops and Conferences

Quantum India, Bangalore 2025
Presented published work on four-qubit systems.

A Hundred Years of Quantum Mechanics — ICTS Bangalore 2025
Attended and participated in discussions on the foundations and future of quantum mechanics.

TQC 2025 — IISc Bangalore 2025
Volunteered, chaired sessions across all days, and developed the official conference website TQC2025.

Other Workshops 2022
IBM–IISc Workshop on Quantum Computing and Hybrid Clouds; The Microsoft Garage Student Innovation Workshop, Workshop 1 & 2

Extracurricular and Volunteer Experience

- **Co-Convener, Carrom Club, IISc** (Nov 2021 – July 2025): Conducted offline events and inter-departmental competitions.
- **Coordinator, Pravega Innovation Summit** (Sep 2022): Organized an innovation competition for registered startups with judges and speakers.
- **Volunteer, Notebook Drive** (2022): Led the stationery-distribution initiative.
- Represented IISc in inter-college and university football and carrom competitions (2022–24).

References

Prof. Sumilan Banerjee

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Prof. C. M. Chandrashekar

Associate Professor
Department of Electronic Systems Engineering
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